Same Motion, Different Scores: Supplementary Material

Anonymous CVPR submission

Paper ID DRAFT

This supplement contains the equivalence proof and verification (Sec. S1), equal-budget and physical controls (Secs. S2 and S3), the full public-output audit (Sec. S4), attack protocols (Sec. S5), reward fine-tuning (Sec. S6), the sign-language experiments (Sec. S7) and statistical details (Sec. S8). Code, configurations, saved outputs and checkpoint hashes accompany the submission.

HumanML3D [1] and the KIT Motion-Language (KIT-ML) dataset [2] pair 3D human motion sequences with text descriptions. We use Guo et al.'s text-to-motion (T2M) evaluator [1] and Text-to-Motion Retrieval (TMR) [3]. R-Precision (R1 for top-1) measures retrieval of a paired caption among candidate descriptions; FID applies the Fréchet distribution distance from image evaluation [4] to motion embeddings [1].

S1. Equivalence and verification

Fiber. For $a \neq 0$ and b not parallel to a, let $u = a/\|a\|$, $b_\perp = b - (u^\top b)u$ and $v = b_\perp/\|b_\perp\|$. Under $a' = \alpha a$, $b' = \beta b + \eta a$ with $\alpha, \beta > 0$, the first vector normalizes to u and $b'_\perp = \beta b_\perp$, so v and $R = [u; v; u \times v]$ are unchanged. Conversely, every d with representative (u, v) is $a = \rho u$, $b = \sigma v + \tau u$ with $\rho, \sigma > 0$, and any two such encodings are related by one recoding. Recodings compose as $(\alpha_2\alpha_1,\ \beta_2\beta_1,\ \beta_2\eta_1 + \eta_2\alpha_1)$, with inverse $(\alpha^{-1}, \beta^{-1}, -\eta/(\alpha\beta))$. Negative scales can flip rows and are excluded.

Canonicalization. $\mathcal{C}(a, b) = (u, v)$ is idempotent and constant on fibers, so $f \circ \mathcal{C}$ is invariant for any deterministic reader f. Its derivative vanishes along $(a, 0)$, $(0, b)$ and $(0, a)$, the three fiber directions.

Numerical checks. On 4,096 random double-precision features, fiber reconstruction error is 4.6×10^{-15}, canonical invariance error 4.9×10^{-15} and directional derivatives are below 3×10^{-14}; our converter matches the official one to 4.5×10^{-16}. For every experiment, saved arrays are checked independently of any score: the maximum rotation discrepancy is below 4×10^{-5} degrees, HumanML3D position channels are bitwise unchanged, sampled forward kinematics differs by at most 9.6×10^{-7}, and all non-rotation channels and padding are bitwise identical. Because tiny floating-point differences can flip a near-tied decoder, hypothesis equality is verified on saved strings rather than inferred.

Table S1. **Equal-budget controls, $\kappa = 3$, mean of 20 draws.** Left: T2M ΔFID; right: TMR ΔR1 in percentage points. Every fixed-fiber T2M interval excludes zero; all intervals and $\kappa = 1$ are in the released tables.

	T2M ΔFID			TMR ΔR1 (pp)		
Block	random	fixed +	fixed −	random	fixed +	fixed −
Fiber	0.0024	0.0823	0.1169	+0.01	−0.10	−0.23
Rotation	0.0035	0.0106	0.0151	+0.02	−0.22	+0.01
Velocity	0.0159	0.0603	0.4760	−0.00	−0.22	−0.19
Contact	0	0	0	−0.24	−0.84	−2.32
Position	0.0183	0.3416	1.1890	+0.03	−0.18	−0.08

S2. Equal-budget block controls

For evaluator r with fixed per-channel standard deviations D_r, each intervention on a clip of L valid frames is rescaled to $\|D_r^{-1}(y - x)\|_2 = \kappa\sqrt{L}$, counting all 263 channels. *Fiber* interventions change only redundant coordinates; *rotation* interventions apply a local SO(3) change and re-encode it canonically; the other blocks receive additive changes to velocities, contacts or positions only. *Random* uses i.i.d. Gaussian directions; *fixed* uses one direction shared by all joints and frames, with both signs. Only fiber interventions keep every represented quantity fixed. The T2M evaluator does not read contacts.

Within the rotation block, the two tested fixed fiber directions change T2M FID by 7.76× and 7.73× as much as the fixed rotation directions of the same sign; at $\kappa = 1$ the ratios are 8.07 and 7.68. These are descriptive ratios for the tested directions, not a universal ordering: random fiber coefficients have smaller effects than random rotation changes,

Table S2. **FID of physical and encoding changes** against the same reconstructed real motion, mean of 20 draws.

Angle	1°	3°	10°	30°
Fixed axis	0.0001	0.0026	0.300	14.44
Per-frame axis	0.350	10.24	38.98	75.17

Recodings (no physical change): fixed ±0.1: 0.059 / 0.083; fixed ±0.35: 1.97 / 3.57; random 0.35: 0.024.

and fixed directions in non-rotation blocks can have larger effects (Tab. S1). What distinguishes the fiber is that it changes no represented quantity, so no response to it is warranted.

S3. Physical calibration

We rotate the 21 non-root joints of real HumanML3D motion by a local angle about a random axis, either fixed through the clip or redrawn per frame, and recompute rotations, positions, velocities and contacts with the native skeleton. All conditions are compared with the same zero-angle reconstruction, whose own FID to the original data is 0.020 (R1 0.451→0.447).

S4. Public-output audit

Protocol. We use the native HumanML3D test loader [1] with its caption, crop and mirroring draws (4,640 entries after dropping the last incomplete batch of 32) and 20 seeds; the real reference is an independent draw. The Human Motion Diffusion Model (MDM) [5], MotionDiffuse [6] and DiP [7] emit 1,024 outputs per draw and the others 4,640; KIT-ML uses 704. Native inference settings are kept: MDM 50 steps (CFG 2.5), MLD [8] DDIM-50 (CFG 7.5), MotionLCM [9] one step, T2M-GPT [10] categorical sampling, T2M with its length estimator, MoMask [11] 18 mask steps, MotionDiffuse 1,000-step DDPM, and DiP with a 20-frame prefix excluded from scoring. Raw and canonical scores share every output, caption, candidate batch and reference. Intervals are paired t_{19}, Bonferroni-adjusted over ten FID/R1 endpoints per study.

Second checkpoint and retrieval. Table S3 repeats the original five releases with the distinct official T2M weights distributed with T2M-GPT. Four FID gaps exclude zero there; MLD's is inconclusive and T2M's changes sign. Retrieval gaps are small with both checkpoints and with TMR (Tab. S4).

Ranking changes. For every pair of releases we compute the per-draw FID and R-Precision difference under raw and under canonical inputs, which is paired because all releases share the same 20 caption draws and reference

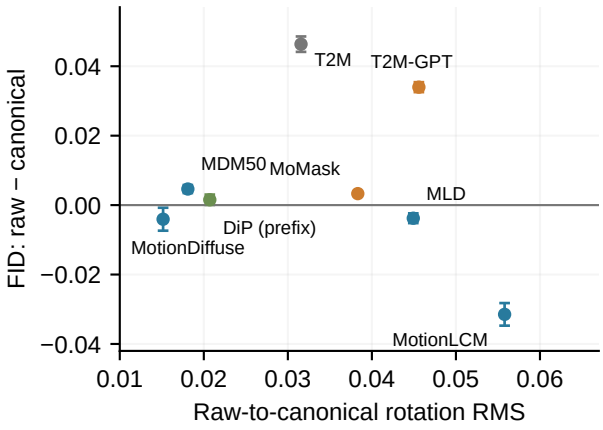


Figure S1. **Gap versus distance from canonical rotations** for the eight HumanML3D releases, with paired intervals. Blue: diffusion/consistency; orange: VQ; gray: VAE; green: prefix-conditioned.

draws, and form t_{19} intervals Bonferroni-adjusted over all pairs and both encodings (56 comparisons for the eight releases, 20 for the five scored by the second checkpoint). With the main checkpoint, the FID ranking of all eight releases is identical before and after canonicalization (Kendall $\tau = 1$). With the second checkpoint, the means of MLD and MotionLCM swap (raw difference $+0.001$, interval $[-0.027, 0.030]$; canonical -0.008, $[-0.039, 0.024]$); the change in their difference, 0.009 $[0.005, 0.013]$, excludes zero, but neither ordering is supported. For R-Precision, MDM and MLD swap in the means with both intervals containing zero. No pair has a supported ordering under one encoding and the opposite supported ordering under the other.

Sign of the gap and model family. Figure S1 plots each release's gap against how far its raw rotations are from canonical. Three of four diffusion or consistency releases have negative gaps and both VQ releases positive ones; the distance itself does not predict the gap (Spearman -0.10 signed and 0.29 absolute; exact permutation $p = 0.84$ and 0.50). Removing rotation non-canonicality therefore does not uniformly favor any model family.

Essential-feature readers. The public MARDM [12] GRU and contrastive readers, which read only the 67 root and position features, are exactly invariant to rotation recoding. Their release rankings differ from that of canonical T2M by two adjacent swaps each (Kendall $\tau = 0.857$ for both).

Published margins. Table 1 of the main paper compares gaps with author-reported 95% margins. These margins de-

Table S3. **Raw minus canonical with two official T2M checkpoints**, 20 draws. Intervals adjust ten endpoints per checkpoint.

Checkpoint	Release	Raw FID	$\mathcal{C}$ FID	FID gap 95% CI	Raw R1	$\mathcal{C}$ R1	R1 gap 95% CI
MDM-linked	MDM	0.4550	0.4504	$[+0.0035, +0.0058]$	0.4115	0.4115	$[-0.0017, +0.0018]$
MDM-linked	MLD	0.4453	0.4491	$[-0.0052, -0.0024]$	0.4118	0.4105	$[+0.0003, +0.0023]$
MDM-linked	MotionLCM	0.5747	0.6062	$[-0.0347, -0.0282]$	0.4374	0.4350	$[+0.0005, +0.0043]$
MDM-linked	T2M-GPT	0.1828	0.1488	$[+0.0325, +0.0354]$	0.4226	0.4240	$[-0.0030, +0.0002]$
MDM-linked	T2M	1.3935	1.3471	$[+0.0441, +0.0486]$	0.3933	0.3930	$[-0.0009, +0.0017]$
T2M-GPT	MDM	0.5129	0.5104	$[+0.0012, +0.0040]$	0.4586	0.4584	$[-0.0015, +0.0019]$
T2M-GPT	MLD	0.4206	0.4216	$[-0.0023, +0.0002]$	0.4669	0.4666	$[-0.0012, +0.0017]$
T2M-GPT	MotionLCM	0.4195	0.4294	$[-0.0138, -0.0060]$	0.5042	0.5026	$[-0.0004, +0.0035]$
T2M-GPT	T2M-GPT	0.1204	0.1115	$[+0.0076, +0.0103]$	0.4909	0.4898	$[-0.0004, +0.0026]$
T2M-GPT	T2M	1.0836	1.0917	$[-0.0111, -0.0051]$	0.4547	0.4540	$[-0.0007, +0.0021]$

Table S4. **TMR retrieval, raw minus canonical**, 20 draws; intervals adjust twelve endpoints.

Release	Raw R1	R1 gap 95% CI
MDM	0.5578	$[-0.0015, +0.0029]$
MLD	0.5848	$[-0.0005, +0.0019]$
MotionLCM	0.6785	$[+0.0013, +0.0043]$
T2M-GPT	0.6703	$[-0.0006, +0.0017]$
T2M	0.5626	$[+0.0003, +0.0028]$

scribe different protocols and are used only for scale. We use MDM's corrected 0.489 ± 0.047, T2M with estimated lengths, MoMask's final KIT table and MotionDiffuse's release preprint.

S5. Attack protocols

Targeted recoding. We sample 1,024 full-caption test clips (40–199 frames) once, with fixed captions and crops, and use them for real motion, MDM outputs and both evaluators. Per-frame, per-joint coefficients start at zero and are optimized for 80 Adam steps (learning rate 0.1, bound 0.35) to minimize embedding distance to the actual or a circularly shifted caption, with evaluator weights frozen. R-Precision averages five fixed 32-candidate permutations. Optimizing through a canonicalizing evaluator leaves every score at its baseline. The main paper uses the T2M checkpoint released with T2M-GPT, whose real-motion scores match the published ones; Tab. S5 repeats the attack with the checkpoint distributed with MDM. Attacks against one checkpoint largely do not transfer to the other (MDM outputs attacked on the MDM-distributed checkpoint reach only 0.512 on the other), as expected for a per-clip attack with gradient access. FID moves in different directions for the two checkpoints, so the attack moves retrieval specifically rather than every metric.

Caption-blind recoding. For 960 fresh MDM outputs whose source clips are disjoint from the targeted study, we compute embedding moments (μ_s, Σ_s) and those of 2,048 training motions (μ_t, Σ_t), and set per-clip targets $\mu_t + (e_i - \mu_s)T$ with the Gaussian transport map [13]

$$T = \Sigma_s^{-1/2}\left(\Sigma_s^{1/2}\Sigma_t\Sigma_s^{1/2}\right)^{1/2}\Sigma_s^{-1/2}, \qquad (1)$$

with eigenvalues floored at $10^{-4}\,\mathrm{tr}(\Sigma)/512$. Coefficients minimize squared distance to these targets for 80 Adam steps. No caption content, test reference or test-time score enters the optimization, and all outputs are saved before reference scoring. Two bounds, 0.1 and 0.35, were fixed in advance; the larger raises FID in every cohort (Tab. S6).

Table S5. **Caption-targeted recoding with both T2M checkpoints**, 1,024 test clips. R1: top-1 R-Precision for the true caption, and for another clip's caption (*other*). FID against the same real test clips. Real-motion R1 is 0.517 (T2M-GPT checkpoint) and 0.462 (MDM checkpoint); the published value is 0.511.

Checkpoint	Motion	R1 before	R1 after	R1 other	FID before→after
T2M-GPT	MDM	0.457	0.997	0.916	0.497→0.393
T2M-GPT	Real	0.517	0.998	0.952	0.000→0.385
MDM	MDM	0.397	0.993	0.788	0.369→1.287
MDM	Real	0.462	0.996	0.839	0.000→1.311

Table S6. **Caption-blind FID**, 960 outputs per cohort. *Source*: evaluator whose gradients were used; *other*: the second official T2M checkpoint, which supplied no gradients. Canonical-gradient controls stay at the original value.

Cohort	Evaluator	Original	$b = 0.1$	$b = 0.35$
MDM seed 42	source	0.384	0.210	1.085
	other	0.408	0.334	0.972
MDM seed 4242	source	0.376	0.176	1.148
	other	0.371	0.303	0.856
MDM seed 7777	source	0.327	0.199	1.429
	other	0.379	0.284	0.779
MLD	source	0.706	0.289	1.611
	other	0.659	0.579	1.345

Query-only recoding. The attacker receives one scalar per query: the FID of its recoded outputs against the 2,048 training-motion embeddings of the primary evaluator. It

sees no gradients, per-clip embeddings, captions or test reference. A (1+1) evolution strategy starts at the identity, proposes Gaussian steps of initial size $0.3b$ (multiplied by 1.5 after a success and 0.9 after a failure) and keeps a proposal only if it lowers the queried FID. It searches either 3 global coefficients or 3 coefficients per joint (63), shared across frames and clips. The best coefficients after a given number of queries are then scored once against the withheld test reference, with the primary evaluator and with the second official checkpoint, which the attacker never queried. Positions, velocities and contacts are bitwise unchanged, and canonical embeddings change by less than 10^{-6}.

With per-joint coefficients and $b = 0.1$, the queried FID falls in every cohort and the fall carries over to the unqueried checkpoint in every cohort (Tab. S7). Most of the gain is reached within 100 queries. The larger bound lowers the queried FID further but overshoots for the other checkpoint, as in the gradient-based caption-blind attack. Three global coefficients are much weaker. R-Precision stays within 0.41–0.44 throughout.

S6. Reward fine-tuning

We fine-tune two generators against a T2M checkpoint and ask whether they move along the redundant coordinates, and we audit one reward-fine-tuned model released by others. Test inference was run only after every setting had been fixed on validation; the dated protocol, including its amendments, is released with the code.

S6.1. MDM

Training. We fine-tune the released MDM [5] (50 sampling steps, classifier-free guidance 2.5) on the 22,414 HumanML3D training clips with whole-motion captions, batch 32, AdamW (learning rate 10^{-5}, weight decay 10^{-4}), gradient clipping at 1 and 2,000 steps; the CLIP text encoder stays frozen. Each step has up to three terms. (i) The usual diffusion x_0 loss on real training motion, computed after canonicalizing the prediction. (ii) The reward, following ReFL [14] and DRaFT [15]: the model samples motions for the batch captions without gradients down to a random step $t_k \in \{0, \dots, 10\}$, predicts x_0 at t_k with gradients, and a frozen T2M checkpoint embeds this prediction, raw or canonicalized depending on the version. The loss is a symmetric InfoNCE over negative Euclidean distances between motion and text embeddings (temperature 1, weight 1), matching how R-Precision ranks captions. (iii) A proximity penalty [16]: the mean squared difference, in normalized units, between the canonicalized x_0 predictions of the fine-tuned and the frozen pretrained model at the same noisy input (weight 100). The control uses term (i) only. Terms (i) and (iii) compare canonical motion, so the redundant coordinates receive gradient only from a raw reward.

Selection. Settings were chosen on 512 validation clips with a criterion fixed in advance: raw R-Precision of the reward evaluator must rise by at least 0.03 while its raw FID stays within 1.5 times the starting value. Without the proximity penalty (diffusion loss on raw or on canonical features, learning rate 10^{-5} or 3×10^{-6}) or with penalty weight 10, reward training raised R-Precision to 0.49–0.58 but failed the FID criterion in every run (one seed each), so the weight was set to 100. At that weight the canonical-reward version qualified (R-Precision 0.420 to 0.477, raw FID 0.416 to 0.285); the raw-reward version exceeded the FID limit only when scored raw (0.660 against a limit of 0.62, canonicalized 0.244), which is the effect under study, so the setting was kept for both. Without the penalty the models moved far from the pretrained one, and every reward run drifted from canonical (off-canonical 0.036–0.065), whichever input the reward read; the penalty prevents this for canonical reward but not for raw reward. In each of the four matched pairs of these earlier runs, the raw-reward version had the larger raw-minus-canonical FID gap (0.31–0.93 against -0.12–0.07). The final protocol, three versions $\times$ three seeds scored once on test with final checkpoints, was fixed before any test inference; the dated protocol is released with the code. The replication with the T2M-GPT checkpoint as reward was then fixed, again before training, with no setting retuned; the no-reward control does not involve an evaluator and is shared.

Test scoring. 2,048 HumanML3D test clips with one caption each, a fixed generation seed and the real test motions as FID reference. R-Precision averages five fixed 32-candidate orders. Intervals use 2,000 paired bootstrap resamples of clips, shared between raw and canonical scoring and averaged over seeds; they do not include seed-to-seed variation, which Tab. S8 shows directly. *Off-canonical* is the RMS difference, over valid frames, between the raw 6D rotation channels and their canonical form.

Results. On validation, the off-canonical distance of every raw-reward run grows steadily, from 0.020 at the start to 0.040–0.046 after 2,000 steps, while the control and canonical-reward runs stay at 0.018–0.019. The test gaps below are raw minus canonical, by the checkpoint named.

MDM checkpoint as reward. Raw reward: FID $+0.467$ [0.278, 0.658] and R1 $+0.008$ [0.002, 0.014] by the reward checkpoint; FID $+0.035$ [-0.035, 0.109] and R1 $+0.002$ [-0.002, 0.006] by the other. Canonical reward: FID -0.044 [-0.076, -0.015] and R1 -0.001 [-0.002, 0.001] by the reward checkpoint. Control: FID $+0.003$ [0.001, 0.005] and R1 $+0.001$ [-0.000, 0.002].

T2M-GPT checkpoint as reward. Raw reward: FID -0.031 [-0.118, 0.058] and R1 $+0.009$ [0.002, 0.016] by the reward checkpoint; FID -0.051 [-0.074, -0.027] and

Table S7. **Query-only recoding**, test FID after 300 queries (relative change from the original). *Queried*: the evaluator that answered the queries; *other*: the second checkpoint, never queried. *Random*: mean of ten random coefficient vectors of the same bound on the queried evaluator.

Search		MDM 42	MDM 4242	MDM 7777	MLD
Original FID (queried / other): 0.384 / 0.408, 0.376 / 0.371, 0.327 / 0.379, 0.706 / 0.659					
Per joint, $b = 0.1$	queried	-20%	-27%	-11%	-57%
	other	-30%	-4%	-22%	-41%
	random	$+2\%$	$+3\%$	$+4\%$	$+4\%$
Per joint, $b = 0.35$	queried	-29%	-44%	-30%	-68%
	other	$+51\%$	$+12\%$	$+57\%$	$+76\%$
	random	$+89\%$	$+74\%$	$+97\%$	$+64\%$
Global, $b = 0.1$	queried	-9%	-11%	-3%	-30%
	other	$+47\%$	$+31\%$	$+28\%$	-18%

Table S8. **Reward fine-tuning, every seed**, 2,048 test captions. Columns give scores by the MDM-distributed and the T2M-GPT-distributed T2M checkpoints. Rows marked MDM or T2M-GPT name the checkpoint that gave the reward; the other one played no part in training.

			MDM checkpoint				T2M-GPT checkpoint		
Model	Seed	Off-canonical	Raw R1	$\mathcal{C}$ R1	Raw FID	$\mathcal{C}$ FID	Raw R1	Raw FID	$\mathcal{C}$ FID
Pretrained	–	0.019	0.415	0.418	0.332	0.327	0.463	0.401	0.400
No reward	42	0.019	0.424	0.423	0.395	0.391	0.480	0.450	0.439
No reward	123	0.019	0.430	0.429	0.398	0.397	0.478	0.470	0.464
No reward	777	0.018	0.427	0.427	0.467	0.464	0.479	0.489	0.481
Canonical reward, MDM	42	0.019	0.490	0.491	0.223	0.261	0.491	0.295	0.286
Canonical reward, MDM	123	0.019	0.491	0.491	0.173	0.211	0.494	0.303	0.296
Canonical reward, MDM	777	0.019	0.488	0.489	0.308	0.364	0.487	0.388	0.379
Raw reward, MDM	42	0.042	0.491	0.491	0.775	0.283	0.488	0.409	0.365
Raw reward, MDM	123	0.043	0.500	0.488	0.656	0.258	0.484	0.416	0.379
Raw reward, MDM	777	0.046	0.496	0.484	0.823	0.312	0.482	0.488	0.464
Canonical reward, T2M-GPT	42	0.018	0.448	0.448	0.280	0.275	0.533	0.151	0.154
Canonical reward, T2M-GPT	123	0.019	0.453	0.453	0.255	0.252	0.532	0.167	0.172
Canonical reward, T2M-GPT	777	0.018	0.448	0.448	0.304	0.299	0.527	0.180	0.184
Raw reward, T2M-GPT	42	0.040	0.448	0.446	0.290	0.333	0.531	0.264	0.211
Raw reward, T2M-GPT	123	0.040	0.451	0.450	0.240	0.292	0.535	0.262	0.277
Raw reward, T2M-GPT	777	0.039	0.445	0.442	0.339	0.397	0.540	0.206	0.338

R1 $+0.002$ [-0.000, 0.005] by the other. Canonical reward: FID -0.004 [-0.007, -0.001] and R1 $+0.001$ [-0.000, 0.002] by the reward checkpoint. Control, by this checkpoint: FID $+0.008$ [0.004, 0.013] and R1 $+0.002$ [0.001, 0.003].

Reading the two checkpoints together. Three findings hold with both rewards: raw reward, and only raw reward, drives the outputs off canonical; it raises the reward checkpoint's R-Precision through the encoding by about 0.01; and it moves FID by more than the control's gap with at least one checkpoint. Size and sign of the FID change differ. With MDM's checkpoint as reward its own FID rises sharply; with T2M-GPT's as reward its own FID gap varies in sign across seeds ($+0.053$, -0.015, -0.132), while MDM's checkpoint, as judge, consistently rates the raw outputs better than their canonical form. The comparison between the two reward versions under that judge (raw scores level, canonical scores separating them in every seed) was not a planned endpoint and is reported as an observation. Like released generators (Sec. S4), canonical-reward models keep small nonzero gaps.

The checkpoint that gave no reward is also less impressed by the fine-tuning itself. After canonicalization it rates canonical-reward models better than the pretrained model (FID 0.32 against 0.40, and 0.28 against 0.33), but raw-reward models no better (0.40 against 0.40, and 0.34 against 0.33); their R-Precision gains over the control are smaller than for the reward checkpoint (0.479 to 0.48–0.49, and 0.427 to 0.45). Raw reward may therefore cost some real quality as well. These are the familiar differences between a proxy reward and an independent judge [17]; the encoding result concerns the raw-minus-canonical gaps of the same outputs and does not depend on them.

S6.2. A second generator: MLD

Training. MLD [8] denoises a 256-dimensional latent that a frozen VAE decodes into HumanML3D features. We fine-tune only the denoiser of the released checkpoint with the same reward as for MDM (MDM's checkpoint, the same InfoNCE loss and weight, reward at a random step among the last 11 of 50), its native DDIM-50 sampler with guidance 7.5, learning rate 10^{-5}, batch 32 and 2,000 steps. Unlike for MDM, both regularizers are MLD's own latent-space terms, which do not single out the redundant coordinates: the noise-prediction loss on VAE latents of real training motion (10% caption dropout), and a penalty on the squared difference between the guided noise predictions of the fine-tuned and frozen denoisers at the reward step.

Selection. The criterion was the one used for MDM. Penalty weights 10, 100 and 1,000 left the reward almost inactive (R-Precision gain at most 0.019 on validation), so none qualified; the grid was then extended to 1 and 0, before any MLD test inference. The fixed rule, the smallest weight at which the canonical-reward version qualifies, selected 0, so unlike MDM the confirmed MLD runs have no proximity penalty. Weight 1, which also qualified, shows the same picture on validation: off-canonical 0.051 (raw reward) and 0.050 (canonical reward), FID gaps -0.009 and -0.005. Three versions $\times$ three seeds were scored once on the same 2,048 test clips as MDM.

Results. Table S9. Both reward versions raise R-Precision equally (from 0.441 to 0.554 and 0.555) and move off canonical form equally (0.061 and 0.060, against 0.044 for the pretrained model and 0.043 for the control). The raw-minus-canonical FID gap by the reward checkpoint changes sign in both: -0.019 [-0.030, -0.010] for the pretrained model and -0.021 [-0.030, -0.012] for the control, against $+0.031$ [0.011, 0.051] for raw and $+0.033$ [0.014, 0.053] for canonical reward; by the other checkpoint, $+0.045$ [0.022, 0.069] and $+0.050$ [0.028, 0.073]. The R-Precision gap of the reward checkpoint is $+0.003$ [0.000, 0.006] for raw and $+0.001$ [-0.002, 0.003] for canonical reward. Unlike MDM, MLD shows no meaningful difference between the two reward inputs: fine-tuning moves its outputs along the redundant coordinates whichever input the reward reads, and the encoding then changes its FID in either case. One possible reason is that MLD reaches the features only through a decoder trained on canonical motion and kept frozen; we did not test this. The absence of a difference is not explained by the missing penalty alone, since validation at weight 1 (one seed) shows it too.

S6.3. A released reward-fine-tuned model

MotionRFT [18] releases an MLD [8] fine-tuned with the authors' own learned reward, together with the base checkpoint it starts from. We generated five native cohorts of 1,024 outputs with each (captions and lengths of the first five MDM audit draws, generation seeds 2101–2105, the native DDIM-50 sampler with guidance 7.5) and scored raw and canonicalized outputs with both T2M checkpoints against the native test reference, with paired t_4 intervals Bonferroni-adjusted over eight endpoints. The off-canonical distance is 0.044–0.046 for the base model and 0.049–0.051 for the fine-tuned one, higher in every cohort. Raw-minus-canonical FID is -0.016 [-0.039, 0.007] (MDM checkpoint) and -0.003 [-0.010, 0.004] (T2M-GPT checkpoint) for the base model, and $+0.036$ [0.028, 0.045] and $+0.028$ [0.008, 0.048] for the fine-tuned model; R-Precision gaps include zero for both. The MLD study above shows the same pattern, a drift off canonical form and an FID gap that turns positive, whether the reward reads raw or canonical input, so we do not attribute it to the authors' reward reading rotations. What it does show is that fine-tuning changed how much this model's score depends on the encoding, a change that canonical evaluation removes.

S7. Sign language

Data and readers. We use the released SignSparK body and hand rotation features [19] (240 per frame: 10 body and 2×15 hand joints) and the public backtranslation readers [20], one Sign Language Transformer [21] per corpus, with native tokenization, beam search and 300-frame cap. Full test sets contain 642 PHOENIX-2014T [22], 2,183 How2Sign [23] and 1,162 CSL-Daily [24] clips; real-motion BLEU is 15.00, 3.48 and 6.89. Left-hand channels, which the released pipeline preprocesses canonically, are unchanged in every attack.

S7.1. Real-input stability controls

We decode the same full How2Sign and CSL-Daily test inputs under canonicalization, Gaussian noise and physical rotation changes, keeping input order, crops, reader weights, vocabularies and beam settings fixed. Every path shares native preprocessing. The reader receives only motion and valid lengths; captions are used only after decoding to compute BLEU. Canonicalization changes 1,438 How2Sign and 464 CSL-Daily hypotheses, while all 642 PHOENIX hypotheses remain unchanged.

Matched input norms. Each clip uses its actual canonicalization displacement $B = \|\mathcal{C}(x) - x\|_2$ over valid frames. Three fixed Gaussian fields on body/right-hand channels are individually rescaled to B, preserving the preprocessed left-hand stream. Their per-clip average is the primary Gaussian

Table S9. **Reward fine-tuning of MLD**, 2,048 test captions, mean of three seeds. R1 and FID by MDM's checkpoint, which gave the reward; the last two columns by the T2M-GPT checkpoint.

		MDM checkpoint (reward)				T2M-GPT checkpoint	
Model	Off-canonical	Raw R1	$\mathcal{C}$ R1	Raw FID	$\mathcal{C}$ FID	Raw FID	$\mathcal{C}$ FID
Pretrained	0.044	0.441	0.441	0.489	0.508	0.367	0.377
No reward	0.043	0.437	0.438	0.767	0.788	0.520	0.533
Canonical reward	0.060	0.555	0.554	0.235	0.202	0.329	0.279
Raw reward	0.061	0.554	0.551	0.199	0.169	0.292	0.247

control; a secondary field perturbs all 240 channels. No field uses captions or evaluator gradients.

Physical controls rotate both original three-vectors together about a per-joint isotropic axis fixed through time. This changes represented rotations while preserving the vectors' norms and inner product; the left hand remains fixed. The matched control solves a per-clip angle to reach the same B: mean angles are $2.623°$ on How2Sign and $5.293°$ on CSL-Daily, with maxima $5.470°$ and $9.756°$. A separate one-degree control has a different input norm. Realized matched-norm errors divided by $\max(B, 1)$ are below 5×10^{-8}.

Measurements and uncertainty. Changed hypotheses compare decoded strings; edit distance uses words on How2Sign and characters with punctuation on CSL-Daily, without reference captions. Normalized edit distance (NED) divides Levenshtein distance by the larger output length, with minimum one; empty outputs remain included. Twenty thousand paired cluster bootstraps resample the 149 official How2Sign video identifiers or CSL-Daily clips, sharing resampling weights across conditions. Primary intervals adjust twenty endpoints (five summarized conditions, two corpora, two measures); canonical-minus-control contrasts adjust a separate eight-endpoint family. These intervals condition on the readers, test inputs and three noise fields rather than estimating model-training uncertainty.

Results. Canonicalization changes more hypotheses and yields larger NED than same-support Gaussian noise in both corpora; all four adjusted paired contrasts exclude zero. Matched physical changes have larger mean effects. Both How2Sign physical contrasts exclude zero; CSL-Daily's changed-fraction contrast includes zero, while its NED contrast does not. The controls establish encoding sensitivity relative to the tested Gaussian fields, alongside sensitivity to physical changes. They do not imply improved semantic accuracy or that small physical changes are linguistically irrelevant. Tables S10 to S12 report every condition, the primary intervals and the paired contrasts.

S7.2. Learned and per-clip recodings

Learned layers. The PHOENIX layer (299k parameters) reads a frozen text-to-motion diffusion model's output and the source sentence, encoded by the generator's frozen text encoder, and predicts bounded coefficients with cross-attention and three dilated temporal convolutions. It is trained for 20 epochs on 7,096 training motions with the reader's caption cross-entropy; generator and reader are frozen and final checkpoints are used. The How2Sign and CSL-Daily layers (770k parameters) read real training motion and the reader's own word embeddings. Table S13 reports ablations: without the source sentence the gain is small, and with the loss computed after canonicalization it vanishes, as Eq. (4) of the main paper predicts.

Per-clip attacks. On 128 real test clips per corpus, 80 Adam steps toward a circularly shifted sentence raise BLEU against that sentence to 85.19 (PHOENIX), 47.34 (How2Sign) and 32.44 (CSL-Daily), while BLEU against the actual sentence drops to 0.00, 0.00 and 1.26; canonical hypotheses are unchanged. A gloss-level CTC recognizer of a different architecture shows the same failure (wrong-target WER 118%→4%).

Detector statistics. Write $s = (\rho - 1, \sigma - 1, \tau)$. The squared distance to the canonical encoding is $E = \|s\|^2$, and the squared feature norm satisfies $N - E = 2(\rho + \sigma) - 2$, so rotating s about $(1, 1, 0)/\sqrt{2}$ preserves both. A per-clip attack restricted to this circle (200 steps) raises PHOENIX BLEU from 6.62 to 10.72 (+4.09, 95% CI [3.04, 5.24]), while a threshold detector on the RMS distance to canonical has AUC 0.500. Fixing the full 2×2 Gram matrix of (a, b) would leave no channel; matching a few statistics is therefore not a general defense, whereas canonicalization closes the whole fiber.

Reader adaptation. Real How2Sign and CSL-Daily features are not canonical. We fine-tune each public reader for five epochs on raw or on canonical training features, with three seeds each and matched batches. Test BLEU is 3.37 raw versus 3.47 canonical on How2Sign (difference $+0.10$, 95% CI $[-0.01, 0.21]$) and 7.24 versus 7.22 on CSL-Daily

Table S10. **Every real-sign stability condition.** Same full native test clips and frozen decoder. Changed hypotheses use words on How2Sign and characters on CSL; normalized edit distance (NED) divides Levenshtein distance by the larger output length, with minimum one. Each Gaussian field and all BLEUs are retained. Same-support noise and physical controls protect the preprocessed left hand.

Corpus	Condition	Changed %	Mean edits	NED	BLEU
How2Sign	Raw	0.0	0.000	0.000	3.4779
How2Sign	Canonical	65.9	8.121	0.397	3.4840
How2Sign	Gaussian support 1	41.2	5.002	0.243	3.5053
How2Sign	Gaussian support 2	43.0	5.169	0.251	3.5494
How2Sign	Gaussian support 3	41.5	4.812	0.235	3.4852
How2Sign	Gaussian all240	37.2	4.556	0.220	3.5763
How2Sign	Physical 1°	45.4	5.208	0.252	3.6018
How2Sign	Physical matched	70.3	8.634	0.426	3.5538
CSL	Raw	0.0	0.000	0.000	6.8907
CSL	Canonical	39.9	5.167	0.304	7.0662
CSL	Gaussian support 1	22.9	2.984	0.173	6.8595
CSL	Gaussian support 2	21.7	2.855	0.165	6.6456
CSL	Gaussian support 3	22.5	2.916	0.167	6.7068
CSL	Gaussian all240	20.5	2.678	0.156	6.9470
CSL	Physical 1°	13.9	1.757	0.102	7.0123
CSL	Physical matched	44.3	5.923	0.346	6.5762

Table S11. **Primary paired stability intervals.** Twenty-endpoint-adjusted intervals, 20,000 cluster bootstraps; official VIDEO_ID on How2Sign, clip on CSL. The Gaussian support mean averages the three fixed fields per clip. Intervals condition on these test inputs, reader weights and noise fields; they do not estimate model-training uncertainty.

Corpus	Condition	Changed %	95% interval	NED	95% interval
How2Sign	Canonical	65.9	[61.7, 69.8]	0.397	[0.366, 0.429]
How2Sign	Gaussian support mean	41.9	[38.9, 45.0]	0.243	[0.223, 0.264]
How2Sign	Gaussian all240	37.2	[33.8, 40.8]	0.220	[0.197, 0.244]
How2Sign	Physical 1°	45.4	[41.9, 49.1]	0.252	[0.228, 0.278]
How2Sign	Physical matched	70.3	[66.6, 74.1]	0.426	[0.396, 0.458]
CSL	Canonical	39.9	[35.5, 44.2]	0.304	[0.269, 0.339]
CSL	Gaussian support mean	22.3	[19.6, 25.2]	0.168	[0.147, 0.191]
CSL	Gaussian all240	20.5	[17.0, 24.2]	0.156	[0.128, 0.186]
CSL	Physical 1°	13.9	[10.8, 17.0]	0.102	[0.078, 0.125]
CSL	Physical matched	44.3	[40.0, 48.7]	0.346	[0.312, 0.382]

$(-0.02, [-0.19, 0.15])$. Canonical readers keep every hypothesis under fresh attacks and under fixed equivalent recodings of all 10,035 test inputs.

S8. Statistics and reproducibility

Motion audits use paired t_{19} intervals over 20 native draws, Bonferroni-adjusted within each stated family. Sign-language intervals use 20,000 paired bootstrap resamples that recompute corpus BLEU from sufficient statistics, clustered by the 149 official How2Sign video identifiers; seed sensitivity additionally resamples the three fitting seeds. Corpus BLEU and WER are recomputed with independent implementations. Clip selections, seeds and bounds were fixed before scoring, and no checkpoint was selected on test scores. Experiments ran with PyTorch 2.11 and CUDA 12.8 on one RTX PRO 6000; exact floating-point decoding can depend on hardware.

Table S12. **Canonicalization minus matched controls.** Separate eight-endpoint-adjusted paired contrasts. Positive changes indicate a larger decoding change from canonicalization; negative changes indicate a larger change from the control. Neither sign establishes semantic accuracy or that physical changes are linguistically irrelevant.

Corpus	Control	Δ changed (pp)	95% interval	Δ NED	95% interval
How2Sign	Gaussian support mean	+24.0	$[+21.0, +26.7]$	+0.154	$[+0.133, +0.174]$
How2Sign	Physical matched	-4.4	$[-7.6, -1.3]$	-0.030	$[-0.053, -0.006]$
CSL	Gaussian support mean	+17.6	$[+13.9, +21.4]$	+0.136	$[+0.106, +0.167]$
CSL	Physical matched	-4.4	$[-9.0, +0.2]$	-0.042	$[-0.080, -0.005]$

Table S13. **Learned-layer ablations**, full test sets. Raw BLEU and gain over no layer with paired 95% intervals; canonical hypotheses are unchanged in every row. How2Sign/CSL controls use one seed.

Corpus	Layer input / loss	Raw	Gain [95% CI]
PHOENIX	text+motion / raw	15.21	8.81 [7.82, 9.87]
	motion only / raw	7.60	1.21 [0.66, 1.76]
	text+motion / canonical	6.26	-0.13 [-0.32, 0.04]
How2Sign	text+motion / raw	6.11	2.64 [2.27, 3.03]
	text+motion / canonical	3.48	0.00 [-0.18, 0.20]
CSL-Daily	text+motion / raw	7.43	0.54 [0.14, 0.97]
	text+motion / canonical	6.79	-0.10 [-0.40, 0.09]

References

[1] Chuan Guo, Shihao Zou, Xinxin Zuo, Sen Wang, Wei Ji, Xingyu Li, and Li Cheng. Generating diverse and natural 3D human motions from text. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5152–5161, 2022. 1, 2

[2] Matthias Plappert, Christian Mandery, and Tamim Asfour. The KIT motion-language dataset. *Big Data*, 4(4):236–252, 2016. 1

[3] Mathis Petrovich, Michael J. Black, and Gül Varol. TMR: Text-to-motion retrieval using contrastive 3D human motion synthesis. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9488–9497, 2023. 1

[4] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local nash equilibrium. In *Advances in Neural Information Processing Systems*, 2017. 1

[5] Guy Tevet, Sigal Raab, Brian Gordon, Yoni Shafir, Daniel Cohen-Or, and Amit H. Bermano. Human motion diffusion model. In *International Conference on Learning Representations*, 2023. 2, 4

[6] Mingyuan Zhang, Zhongang Cai, Liang Pan, Fangzhou Hong, Xinying Guo, Lei Yang, and Ziwei Liu. MotionDiffuse: Text-driven human motion generation with diffusion model. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(6):4115–4128, 2024. 2

[7] Guy Tevet, Sigal Raab, Setareh Cohan, Daniele Reda, Zhengyi Luo, Xue Bin Peng, Amit H. Bermano, and Michiel van de Panne. CLoSD: Closing the loop between simulation and diffusion for multi-task character control. In *International Conference on Learning Representations*, 2025. 2

[8] Xin Chen, Biao Jiang, Wen Liu, Zilong Huang, Bin Fu, Tao Chen, and Gang Yu. Executing your commands via motion diffusion in latent space. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 18000–18010, 2023. 2, 6

[9] Wenxun Dai, Ling-Hao Chen, Jingbo Wang, Jinpeng Liu, Bo Dai, and Yansong Tang. MotionLCM: Real-time controllable motion generation via latent consistency model. In *Computer Vision – ECCV 2024*, pages 390–408, 2025. 2

[10] Jianrong Zhang, Yangsong Zhang, Xiaodong Cun, Yong Zhang, Hongwei Zhao, Hongtao Lu, Xi Shen, and Ying Shan. Generating human motion from textual descriptions with discrete representations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14730–14740, 2023. 2

[11] Chuan Guo, Yuxuan Mu, Muhammad Gohar Javed, Sen Wang, and Li Cheng. MoMask: Generative masked modeling of 3D human motions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1900–1910, 2024. 2

[12] Zichong Meng, Yiming Xie, Xiaogang Peng, Zeyu Han, and Huaizu Jiang. Rethinking diffusion for text-driven human motion generation: Redundant representations, evaluation, and masked autoregression. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 27859–27871, 2025. 2

[13] Rajendra Bhatia, Tanvi Jain, and Yongdo Lim. On the Bures–Wasserstein distance between positive definite matrices. *Expositiones Mathematicae*, 37(2):165–191, 2019. 3

[14] Jiazheng Xu, Xiao Liu, Yuchen Wu, Yuxuan Tong, Qinkai Li, Ming Ding, Jie Tang, and Yuxiao Dong. ImageReward: Learning and evaluating human preferences for text-to-image generation. In *Advances in Neural Information Processing Systems*, 2023. 4

[15] Kevin Clark, Paul Vicol, Kevin Swersky, and David Fleet. Directly fine-tuning diffusion models on differentiable rewards. In *International Conference on Learning Representations*, pages 4793–4822, 2024. 4

[16] Ying Fan, Olivia Watkins, Yuqing Du, Hao Liu, Moonkyung Ryu, Craig Boutilier, Pieter Abbeel, Mohammad Ghavamzadeh, Kangwook Lee, and Kimin Lee. DPOK: Reinforcement learning for fine-tuning text-to-image diffusion models. In *Advances in Neural Information Processing Systems*, 2023. 4

[17] Leo Gao, John Schulman, and Jacob Hilton. Scaling laws for reward model overoptimization. In *International Conference on Machine Learning*, pages 10835–10866, 2023. 5

[18] Xiaofeng Tan, Wanjiang Weng, Hongsong Wang, Fang Zhao, Xin Geng, and Liang Wang. MotionRFT: Unified reinforcement fine-tuning for text-to-motion generation, 2026. 6

[19] Jianhe Low, Alexandre Symeonidis-Herzig, Maksym Ivashechkin, Ozge Mercanoglu Sincan, and Richard Bowden. SignSparK: Efficient multilingual sign language production via sparse keyframe learning. In *European Conference on Computer Vision*, 2026. 6

[20] LionelLow. SignSparK-BT: Public back-translation checkpoints and model card. Hugging Face, 2026. Audited release trained on re-optimized poses; model revision 0ced459f6c24d453893809eeca87e9ec80b70198 for CSL-Daily. 6

[21] Necati Cihan Camgoz, Oscar Koller, Simon Hadfield, and Richard Bowden. Sign language transformers: Joint end-to-end sign language recognition and translation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10023–10033, 2020. 6

[22] Necati Cihan Camgoz, Simon Hadfield, Oscar Koller, Hermann Ney, and Richard Bowden. Neural sign language translation. In *IEEE Conference on Computer Vision and Pattern Recognition*, pages 7784–7793, 2018. 6

[23] Amanda Duarte, Shruti Palaskar, Lucas Ventura, Deepti Ghadiyaram, Kenneth DeHaan, Florian Metze, Jordi Torres, and Xavier Giro i Nieto. How2Sign: A large-scale multimodal dataset for continuous american sign language. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2735–2744, 2021. 6

[24] Hao Zhou, Wengang Zhou, Weizhen Qi, Junfu Pu, and Houqiang Li. Improving sign language translation with monolingual data by sign back-translation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1316–1325, 2021. 6